
WEAKLY SUPERVISED 3D U-NET FOR ANEURYSM DETECTION

A PREPRINT



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December 5, 2025

ABSTRACT

This paper aims to investigate the use of 3D convolutional neural networks, more specifically 3D U-Nets, for biomedical image segmentation, focusing on the identification of brain aneurysms CT imaging data. The proposed model extends the original U-Net architecture by incorporating attention gates, and a multi-task learning framework including auxiliary classification and regression heads. Weak supervision is implemented via Gaussian heatmap targets trained with a curriculum Dice loss and background-aware regularization. Additionally, this work addresses the typical challenges encountered when working with 3DCNNs, such as preprocessing and working with large amounts of volumetric data, class imbalances, and the amount of resources necessary to train the network.

Keywords Machine Learning · Deep Learning · 3D-CNN · 3D U-Net · CTA · Auxiliary Classification · Attention U-Net · Patch Based Training · Gaussian Heatmap · Semimetric Dice Loss · Curriculum Learning · Multi-task Learning

1 Introduction

Intracranial aneurysms (IAs) are abnormal focal dilations of cerebral blood vessels that, if ruptured, can have fatal consequences. In clinical practice, patients in the early stages generally have no obvious symptoms, making it very easy to miss the diagnosis. It is one of the most common causes of cerebrovascular accident, affecting an estimated 3% - 5% of the general population [1] and up to 50% of cases are identified only after rupture. Once the rupture happens it will cause subarachnoid hemorrhage, which has the mortality rate of 40% and a disability rate of more than 30% [2]. Thus, the development of a system for early identification could be critical in guiding treatment and preventing rupture.

This paper aims to create an aneurysm detection model, which can solve the task of segmentation, classification and localization, using only weak supervisory signals derived from aneurysm coordinates, rather than full voxel-level annotations. However, this comes with its own set of challenges. The medical images overall come from different scanners, with variation in noise levels, voxel spacing, scanning orientation and more, introducing significant variability the model must learn to generalize over. Additionally, aneurysms vary widely in size, shape and location within the brain, with aneurysms as small as 5mm or less. The minuscule size of the aneurysm creates quite a strong class imbalance. Lastly, natural anatomical variation that exists between patients in vessel morphology can obscure or mimic the appearance of an aneurysm leading to the common problem of false positives.

In order to accomplish the proposed goal, the paper employs a variety of strategies inspired by multiple scientific papers. The final model is an improvement upon the classic 3D U-Net architecture, complete with an attention mechanism, classification and regression head. The training pipeline incorporates patch-based sampling with a controlled positive-to-negative ratio, spherical masks and Gaussian heatmaps for soft supervision, and curriculum learning to gradually introduce task complexity. This produces the final model which achieves upon inference of 400 total patients an F1 score of 71.9%, 89.7% recall, and a predicted peak on segmentation of 2mm away from the true target.

2 Related work

Intracranial aneurysm detection and segmentation has been the focus of substantial research in recent years, especially in deep learning and 3D medical image processing. Existing methods can be broadly categorized into segmentation-based deep learning, multi-task pipelines, and clinical validation studies.

A 3D multi-task U-Net that combines segmentation and classification heads for intracranial aneurysms was proposed by [3], using vessel segmentation mask, emphasizing the value of a patch-based approach to handle large volumes. Their multi-task, patch-based framework motivated the choice to explore similar strategies for handling large 3D scans efficiently. Studies relying only on coordinate information introduced two important ideas. One of them [4] generated a synthetic 3D spherical mask centered at each annotated coordinate, turning point-level labels into an approximate segmentation target. The other [5], inspired by nnLandmark, replaced the sphere with a Gaussian heatmap around each landmark coordinate, offering a softer, probabilistic target that better accommodates spatial uncertainty. Even though they used MSE for the loss, [6] proposed a new Dice Loss designed specifically for soft labels, showing that it leads to more stable optimization and improved Dice performance across several medical segmentation benchmarks.

Building upon these insights, this work employs a multimodal 3D U-Net with attention mechanisms and a recently introduced spherical Gaussian heatmap representation for aneurysm localization. The main contributions of the approach can be summarized in three parts. First, the extension of the Semimetric Dice Loss from previous work [6] by adding a curriculum strategy tailored to Gaussian heatmaps. This curriculum operates in two ways: applying a threshold to the Gaussian heatmap by initially setting all probabilities below a value α to zero and progressively lowering α during training, and reducing the standard deviation of the gaussian distribution to make the heatmap more compact as the network improves. Second, this work adopts a multi-task framework with uncertainty-based loss weighting [7], allowing the model to learn segmentation, classification, and coordinate regression in a balanced and stable manner. Finally, a background-aware constraint penalizing high probabilities outside the spherical region is used, helping the model reduce false positives and improve spatial accuracy. Together with a patch-based sampling strategy for memory efficiency, these components form a robust framework for detecting small intracranial aneurysms in large 3D volumes.

3 Methods

3.1 Dataset

The dataset was recovered from the last "RSNA Intracranial Aneurysm Detection" challenge in kaggle located in [8]. It is distributed across two CSV files and a directory containing the full imaging series for each patient. The file `train.csv` includes the patient identifier, a binary flag indicating the presence of an aneurysm, and 13 binary columns describing the anatomical region of the aneurysm for positive cases. The file `train_localizers.csv` contains, for each aneurysm-positive patient, one set of 3D coordinates for each aneurysm present, with each one annotated only by a single voxel marking its approximate center.

In total, the dataset includes 4348 observations acquired at 18 medical sites across five continents. All cases were manually annotated by expert radiologists. The data contain four imaging modalities: CTA, MRA, MRI T1, and MRI T2. Because the preprocessing requirements, value ranges, and acquisition protocols vary substantially across modalities, the dataset was restricted to CTA scans, which are more standardized and homogeneous. Additionally, patients with inconsistent slice orientations or missing slices were removed. After this filtering step, 1482 patients remained, of which 779 had aneurysms and 702 were healthy, resulting in a balanced distribution with respect to the target label. A detailed overview of the resulting dataset is presented in Table 1.

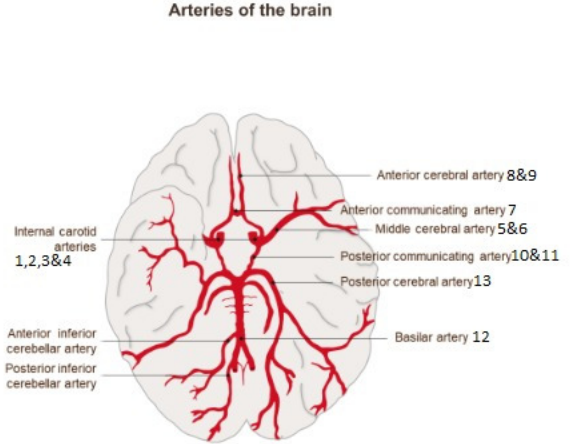
As illustrated in Table 1, the distribution of aneurysm locations is highly imbalanced, with most cases occurring in regions around the internal carotid and middle cerebral arteries. This concentration of cases can bias the model toward overpredicting aneurysms in these regions if not properly accounted for during training.

3.2 Data preprocessing and augmentation

Following the preprocessing pipeline described in [9], the full series of slices for each patient were stacked into a 3D volume and resampled to an isotropic spacing of 1 mm. Using the Hounsfield Unit (HU) ranges characteristic of CTA scans, the head region was isolated by cropping, ensuring that the model focuses solely on relevant anatomical structures. All volumes were then normalized using z-score normalization [10] to reduce inter-scan intensity variability and changed from fp32 to fp16 for memory efficiency. The resampling and cropping transformations were applied consistently to the aneurysm coordinates to maintain spatial correspondence.

Table 1: Dataset description

Observations	1481
Patients with aneurysm	779
Healthy patients	702
Aneurysm counts by vascular region	
Region 1	28
Region 2	41
Region 3	123
Region 4	93
Region 5	99
Region 6	130
Region 7	164
Region 8	26
Region 9	27
Region 10	54
Region 11	65
Region 12	50
Region 13	49
Total aneurysms	949



Because the volumetric dimensions vary substantially across patients—and training a 3D U-Net directly on large volumes is computationally expensive—a patch-based strategy was adopted. Cubic patches of size $64 \times 64 \times 64$ were extracted from each volume, following the sampling strategy described in [3]. To address class imbalance, patches containing aneurysms were oversampled, maintaining an approximate ratio of 1:2 between aneurysm and healthy patches. For aneurysm-containing patches, random local offsets around the annotated coordinates ensured coverage of different anatomical contexts and prevented the model from overfitting to a fixed relative neighborhood.

For segmentation training, the relative aneurysm coordinates were used to generate a 3D Gaussian mask centered at the aneurysm location. Although a binary spherical mask was also evaluated, the Gaussian representation proved more suitable for guiding the network toward the aneurysm region. Data augmentation was applied during training to improve generalization and reduce overfitting. In addition to the positive-patch variation introduced earlier through random local offsets, random rotations on-the-fly to further diversify the spatial appearance of both aneurysm and non-aneurysm regions were used.

3.3 Network architecture

In the context of medical image segmentation, the one architecture that has proven itself to be consistently outperforming other methods, is the U-Net, originally presented in the paper published by Ronneberger et al [11]. Ever since it has been widely used in various medical image segmentation tasks, such as brain tumor segmentation, lung segmentation, and cell segmentation. Furthermore, since the data presented is all volumetric, the 3D U-Net architecture would be the perfect fit for the problem. The architecture presented in this paper follows the original U-Net structure with multiple improvements and adjustments which better fit the proposed goal, the final version is presented in Figure 1

The base architecture can typically be divided into 3 parts: the encoder/contracting path, bottleneck block, and the decoder/expanding path. The encoder of the network captures the context of the input through successive convolutional and pooling layers, gradually doubling the number of feature channels at each downsampling step, thus the network learning increasingly more complex features. Conversely, the decoder restores spatial resolution using upsampling and includes concatenation with corresponding feature map from the encoder. The upsampling is done using trilinear interpolation in order to make sure there is no checkerboard artifact when reconstructing the volume. The bottleneck consists of two convolutions with no pooling, and it connects the encoder and decoder at the deepest part of the network. Lastly, the final 1×1 convolution maps each feature vector to the desired number of classes, producing the segmentation map.

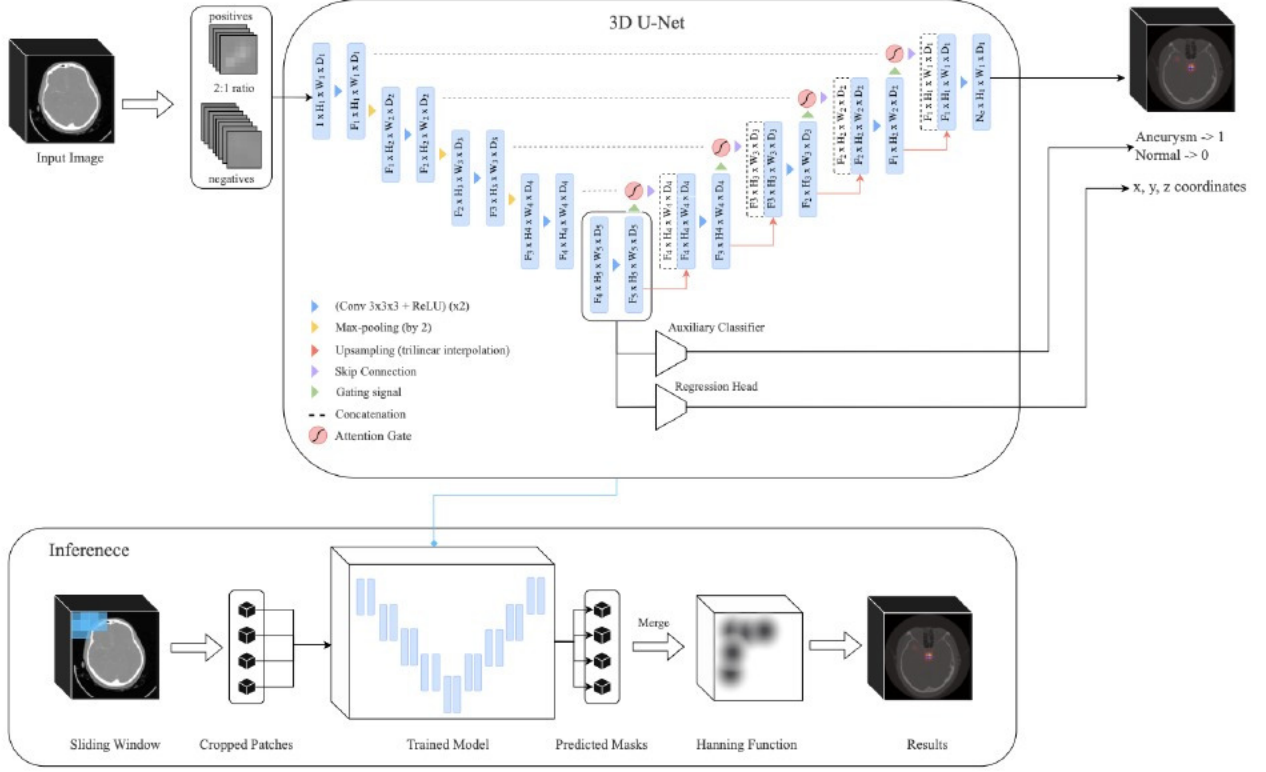


Figure 1: Network Architecture

The model’s encoder-decoder structure, coupled with the skip connections, enables the network to both capture the fine details for precise localization, as well as all encompassing contextual information. The improvements that were added on top of the original architecture include flexible normalization, optional classification and regression heads, and attention mechanisms.

The auxiliary classifier and the regression head are located in the deepest part of the network, at the level of the bottleneck. An auxiliary classifier can improve the performance of the classification within the network by providing an additional supervision signal. It stabilizes training and reduces the risk of vanishing gradients [3]. The regression head, in a similar fashion, predicts the spatial coordinates within the volume which helps to guide the feature extraction process resulting in more effective processing of spatial information [5].

The final enhancement is a fairly simple, but a trusted method of improving a U-Net’s performance - attention gates. Attention gates within a model function similar to a filter, highlighting the important features while suppressing irrelevant ones. They are added at each level of the skip connections in the network, right before the concatenation operation, and have been shown to consistently improve model sensitivity and prediction accuracy [12].

The full preprocessed volume is first used to generate positive and negative patches, these are run through the described network, which produces three outputs: a segmentation prediction, a classification score and coordinate predictions. During training, the ground-truth locations are converted into 3D Gaussian heatmaps, which serve as soft targets for the regression head. These heatmaps provide smooth spatial supervision, helping the model learn not only the exact location point but also its surrounding spatial context. All task specific losses, such as segmentation, classification, and regression loss computed from the Gaussian heatmaps, are summed into a weighted multi-task loss that guides learning. The weights are gradually adjusted throughout training through backpropagation and multiple metrics are generated as well in order to monitor and evaluate the model progress.

For inference, the pipeline operates on the full 3D volume using a sliding-window approach. For each patch generated, only the segmentation logits are used, and their sigmoid-activated probabilities are blended back into the global probability map using a Hann-window weighting to ensure smooth transitions between patches. After all patches are processed, the aggregated probability map is normalized by the accumulated weights and analyzed.

3.4 Loss function

The objective of the multi-task loss function presented below follows four main tasks: (i) segmentation of the synthetic Gaussian blobs, (ii) classification of aneurysm presence, (iii) regression of the aneurysm coordinates, and (iv) suppression of false positives.

Beginning with the segmentation objective, the Semimetric Dice loss for soft labels proposed in [6] is extended by incorporating a curriculum-learning mechanism [13]. Dice loss was selected at first because, combined with the U-net architecture, the model can learn the brain structure. Moreover, the motivation to modify the Dice Loss is that early training makes it difficult for the network to distinguish the Gaussian core from its tail and background; the curriculum threshold helps it learn this separation progressively. Let $\Omega \subset \mathbb{R}^3$ be the spatial domain of the image, $X \in [0, 1]^\Omega$ be the model prediction, and $Y \in [0, 1]^\Omega$ be the ground-truth Gaussian heatmap. A curriculum parameter $\alpha \in [0, \alpha_{max}]$ defines two Masks: a core and a background region.

$$M_{core, \alpha} = Y \cdot (Y \geq \alpha), \quad M_{background, \alpha} = 1 - M_{core} \quad (1)$$

$$\mathcal{L}_{CSDL}(X, Y, \alpha, \lambda_{tail}) = 1 - \frac{2\langle X \odot M_{core}, Y \odot M_{core} \rangle + \epsilon}{2\langle X \odot M_{core}, Y \odot M_{core} \rangle + \mathcal{D}_{core}(X, Y) + \lambda_{tail} * \mathcal{D}_{background}(X, \alpha) + \epsilon} \quad (2)$$

where $\odot$ denotes element-wise multiplication, and $\epsilon > 0$ stabilizes division.

The disagreement terms quantify deviations within the core and tail regions:

$$\mathcal{D}_{core}(X, Y) = \|(X - Y) \odot M_{core}\|_1 \quad (3)$$

$$\mathcal{D}_{background}(X, \alpha) = \sum_{\mathbf{u} \in \Omega} M_{background}(\mathbf{u}) \cdot \max(0, X(\mathbf{u}) - \alpha) \quad (4)$$

To further mitigate confident false positives outside the Gaussian blob, a max-penalty term is used, scaled by a factor of λ , resulting in the final segmentation objective.

$$\mathcal{L}_{seg}(X, Y; \alpha, \lambda_{max}, \lambda_{tail}) = \mathcal{L}_{CSDL}(X, Y, \alpha, \lambda_{tail}) + \lambda_{max} \max_{\mathbf{u} \in \Omega} (X(\mathbf{u}) M_{background, \alpha}(\mathbf{u})). \quad (5)$$

In addition to segmentation, the model includes a classification head, supervised using a weighted BCEWithLogitsLoss to address the imbalance between positive and negative patches. Furthermore, the regression head responsible for predicting aneurysm coordinates is trained using Mean Squared Error (MSE). Finally, a background-suppression loss derived from z-score normalization is incorporated, where the values outside the brain are below 0. This encourages the model to down-weight predictions in anatomical background:

$$M_{outside\ brain} = (I(\mathbf{u}) < 0) M_{background, \alpha}, \quad \mathcal{L}_{outside\ brain} = \frac{\sum_{\mathbf{u} \in \Omega} X(\mathbf{u}) M_{outside\ brain}}{\sum_{\mathbf{u} \in \Omega} M_{outside\ brain} + \epsilon}. \quad (6)$$

Balancing multiple objectives is non-trivial, as each task has different scales and degrees of noise. Therefore the uncertainty-guided weighting strategy of [7] is a key component of the solution, which introduces a learnable log-variance ($s_t = \log \sigma_t^2$) for each task t . tasks with higher uncertainty (larger σ_t) receive lower weight, while more reliable tasks receive higher weight. The final total loss is:

$$\mathcal{L}_{total} = \sum_{t \in \{CSDL, BCEWithLogitsLoss, MSE\}} \left(\frac{1}{2} \exp(-s_t) \mathcal{L}_t + \frac{1}{2} s_t \right) + \lambda \mathcal{L}_{outside\ brain} \quad (7)$$

3.5 Training details

Training was performed on 11,931 3D patches of size 64^3 , split into 70% for training and 30% for validation, using a batch size of 20 with two-step gradient accumulation, running in fp16 [14] via Accelerate and DeepSpeed (ZeRO Stage 2 [15]), with TorchDynamo-Inductor in reduced-overhead mode using dynamic and full-graph regional compilation.

The network is a five-level attention U-Net with 32 initial filters, with classification and regression fully connected heads at the end of the encoder, group-4 normalization, trilinear upsampling, same padding, and dropout of 0.2. Optimization uses Adam8bit with $\beta = 0.9/0.999$, learning rate 10^{-4} , weight decay 10^{-3} , cosine annealing to 10^{-6} over 40 epochs, and gradient clipping at 5.

Targets are Gaussian heatmaps with an initial $\sigma = 5$, which decays (cosine) to $\sigma = 0.01$ between epochs 20 and 30. The segmentation loss $\mathcal{L}_{\text{seg}}$ (Eq. 5) uses a curriculum threshold α that also follows a cosine decay from 0.3 to 0.01 over the first 20 epochs. Loss weighting includes $\lambda_{\text{tail}} = 0.1$, $\lambda_{\text{max}} = 0.1$, and a background-suppression term $\lambda_{\text{outside brain}} = 0.1$ while the other weights are learned by multi-task uncertainty. Training was executed on an NVIDIA A10G GPU with 24 GB of memory on an AWS g5.4xlarge instance and completed in approximately 10 hours.

4 Results

To address the performance of our solution, different variants of the network were considered, disabling one or more of the features to see their impact on the training and the results. That way, 8 different models were tested for comparison, disabling attention, background loss, classification and regression heads, and curriculum learning, as well as keeping the model with all the features but varying the proportion of positives to negatives.

The first direct comparison between all the models is their direct segmentation outputs, along with the attention maps of the network in the last layers

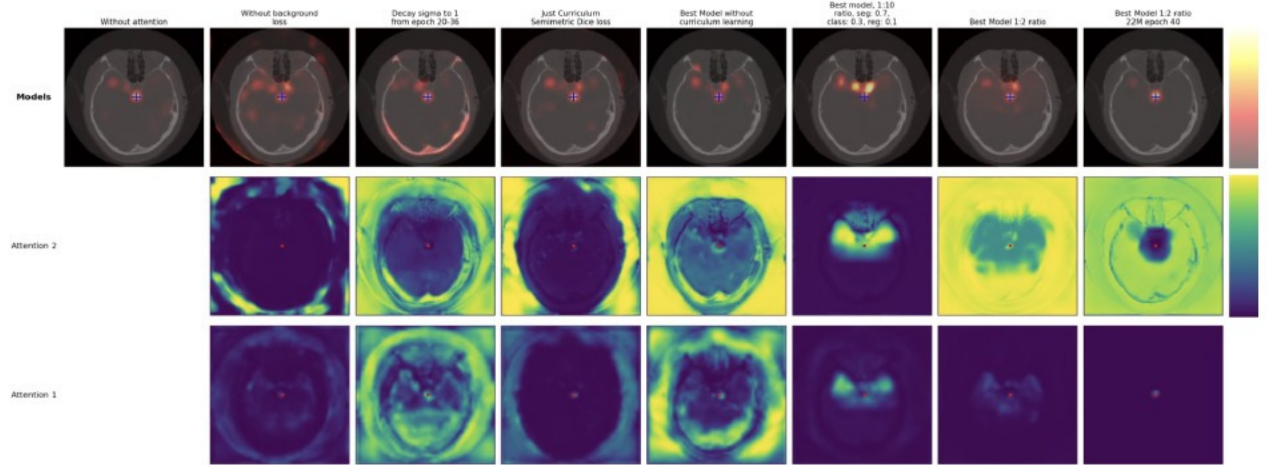


Figure 2: Attention maps

For the positive samples, all the models found the aneurysm in the correct voxel. However, some of them showed a lot of noise elsewhere; the one without background penalty was not capable of suppressing the background of the image, and the models with decaying sigma and just semimetric loss had a lot of false positives in the same slice. Adding both a background loss and multiple heads reduced false positives and improved focus. Increasing the capacity of the model by adding one downsampling and upsampling block to the Unet reduced probabilities in irrelevant areas.

Regarding attention maps, the last 2 maps were analyzed, as early layers looked similar for all of the models. Most models exhibited an inverted attention map in the last layer, acting more like a suppression mask to ignore background regions. The 1:10 model with fewer positive samples focused more on the aneurysm locations, rather than on suppressing the background and the noise. In the second layer of attention maps, most models acted as a spatial attention, focusing on the location of the aneurysm, which is clearer in the best models.

The results and attention maps for a healthy patient are as follows:

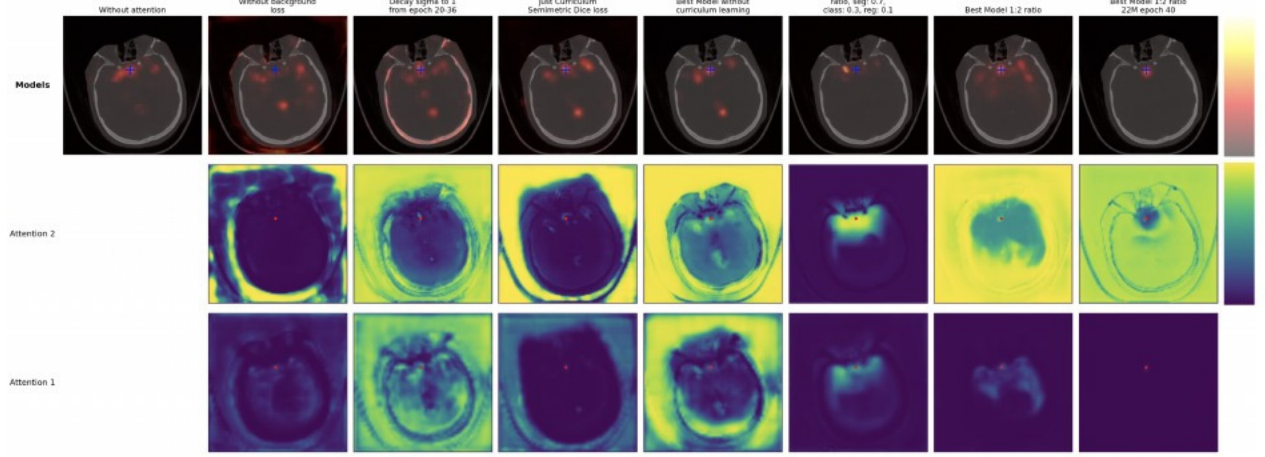


Figure 3: Attention maps for false positive

All the networks predict the presence of an aneurysm for a healthy patient, reflecting a bias in the original dataset, where most of the aneurysms are located in the same location. As the model becomes more complex and acquires more representational power, the false positive probabilities are reduced.

Metrics were defined for fair comparison across heterogeneous models, as standard metrics like IoU or loss function values were unsuitable. The metrics and their raw results are as follows:

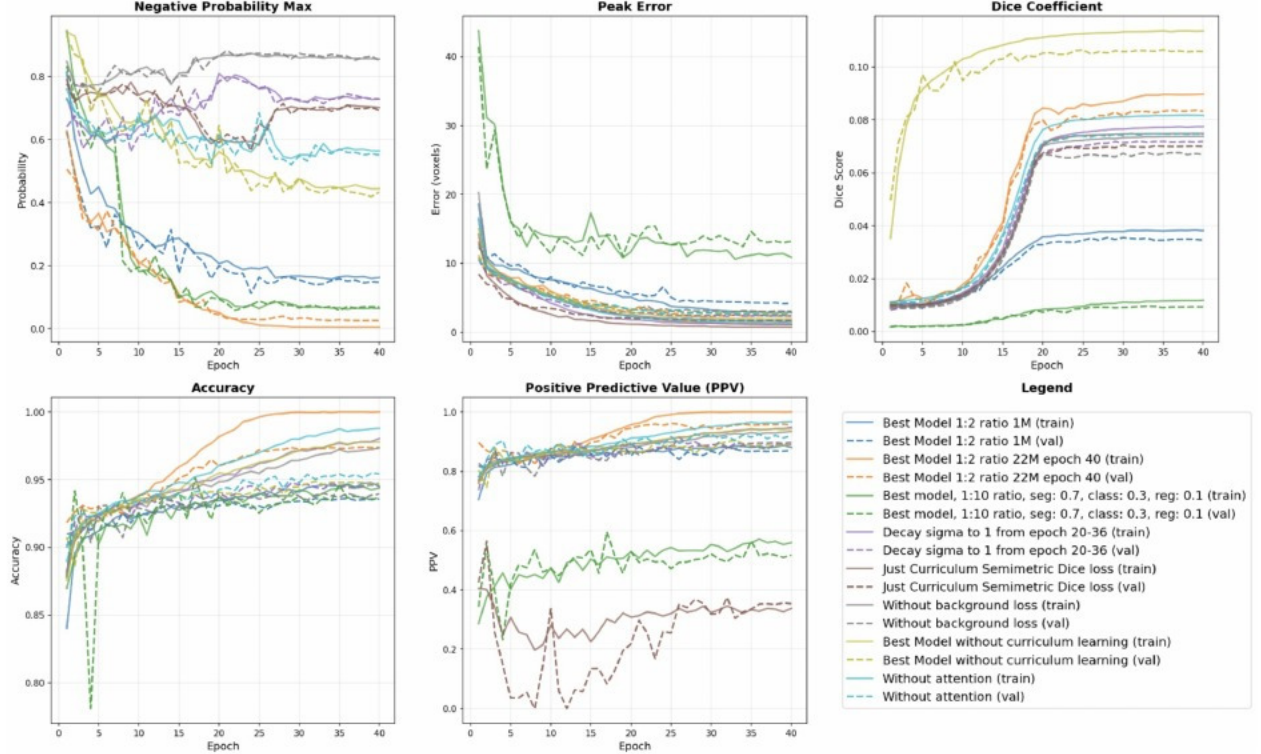


Figure 4: Metric comparison

- **Negative Probability Max:** This is the average maximum probability output by the segmentation head when the input for this is a negative patch. Here, Models without the background penalty had higher false positives.

- **Peak error:** The peak error represents the distance between the maximum predicted probability by the model and the aneurysm coordinates. This metric gives a value of how precise is the prediction of the segmentation and is very useful in scenarios like this, where an artificial mask is generated around the point of interest. Most models converge to a peak error near zero, except the one with a class imbalance of 1:10, which was biased toward negative predictions.
- **Dice coefficient:** The soft DICE coefficient is calculated between the output of the model and the gaussian mask. The best model without curriculum learning achieved high DICE from the first epoch because it didn't allow the model to predict probabilities outside the gaussian mask during the first epochs.
- **Accuracy:** This plot contains the accuracy per patch. The largest model overfitted, reaching 1 on training and 0.96 on validation; other models showed similar trends. Models without curriculum learning and only Dice Loss performed poorly.
- **Positive Predictive Value (PPV):** The PPV is a measure of how many positive predictions were actually positive and is useful to assess the proportion of false positives outputted by the network. Most models overfitted, where the best model has a PPV of 1 on training but 0.9 on validation. The 1:10 ratio model had a low PPV (0.5), reflecting class imbalance.

Finally, full-volume inference considered outputs from all three heads. Observations were classified as positive if thresholds from classification and segmentation heads were exceeded and the regression output matched the segmentation maximum within a threshold.

Doing inference on 400 total patients, the best model achieved 71.9% F1 score, 89% recall, 60% recall, and 60% accuracy. Compared with a study [16] that evaluates the performance of 120 clinicians on the same task, they achieved 73% accuracy and 76% F1 score, having a difference between the two of 4.1% F1 score and 13% accuracy, respectively. Even though our best model has many false positives, it succeeded in having small false negatives, which is important in the medical field.

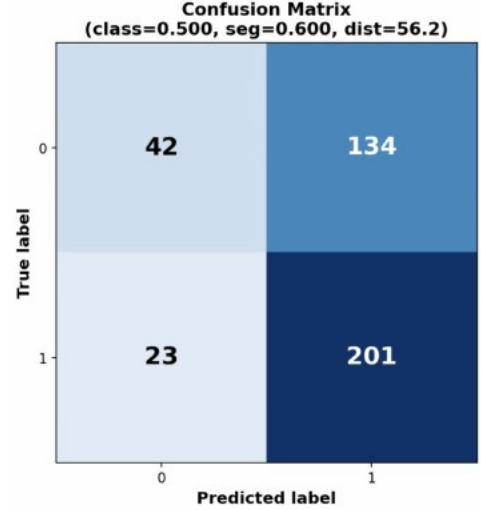


Figure 5: Confusion matrix

5 Discussion

From the segmentation and metrics results, it is evident that the inclusion of more parameters, a background penalty term in the loss, curriculum learning, and an appropriate choice of the ratio of positive and negative patches are key components of the solution, leading together to a network capable of doing segmentation and classification of aneurysms.

However, the lack of a vessel segmentation mask that is present in more state-of-the-art solutions was determinant in bounding the region of interest and training a model that focused on the right structures from the beginning. On the other hand, the approach for generating the patches, specifically the negative ones, could be improved; the current approach is too naive and is prone to generating patches of unrelated regions for the task. Instead, a possible improvement would be to create patches with the false positives of the training set, assessing specific regions where the network has difficulties.

Finally, a pain point during the training process was the lack of metrics that allowed for a fair comparison, since the architectures and loss functions were so heterogeneous; hence, some metrics were defined, but it is necessary to address their robustness. The full implementation and process can be accessed in the github repository [17].

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